

AL-HIFAZAT MILU

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RESEARCH INTERESTS

- Thermal System Design
- Robotics & Automation
- Vehicle Dynamics and Control Systems
- Micro & Nano-materials Technology
- Additive Materials and Advanced Manufacturing
- Energy Systems Utilization

EDUCATION

B.Sc. in Mechanical Engineering

Feb 2016–Mar 2021

Rajshahi University of Engineering and Technology (RUET), Rajshahi, Bangladesh

CGPA: 2.88 out of 4.00 (CGPA of last 3 semesters/61.25 credits: 3.25 out of 4.00)

- B.Sc. Thesis: “*Fabrication and Performance Analysis of an IoT-Based Smart Solar PV-Panel Tracking System*,” under supervision of Dr. Nirendra Nath Mustafi
- Departmental Project: “*7E analysis of a modified insulated stepped pyramid solar still*,” under supervision of Md. Bakhtierkhalzi

WORK EXPERIENCE

Refrigeration Systems Design & Control Engineer

Nov 2021–Present

Research & Innovation, Refrigerator, Walton, Gazipur, Bangladesh

- Initiated new projects focused on **performance improvement** and **cost-effective design** by applying new components and addressing market-driven requirements and designed optimized cooling circuits accordingly.
- Provided the complete **IoT and control-system logic** requirements for refrigerator platforms to the electrical and electronics development team, enabling implementation on **STM32, ARM, and AVR** microcontroller architectures using **STM32CubeIDE, Keil uVision, and Atmel Studio**.
- Prepared performance, reliability, and safety **test protocols**, conducted **development tests** and performed **feasibility analyses** using computational and analyzing software to ensure compliance with refrigeration cycle quality, efficiency, and longevity standards.
- **Designed** cooling tube bending **dies and moldable component parts** using SolidWorks, ensuring manufacturability and structural accuracy.
- **Measured thermal conductivity, thermal intensity or noise level** of the refrigerator samples or its parts by using Hukseflux, Thermal Camera, Noisecam machines respectively.
- Implemented **data-driven solutions** and prepared **technical documentation** including engineering drawings, validation reports, and production and after-sales service instructions by analyzing manufacturing and market issues.

Project Manager

Jul 2021–Sep 2021

Operation team, Rynco Engineering LTD, Gazipur, Bangladesh

- **Led a team** in the installation of central fire pumps, piping systems, and fire sprinklers, employing advanced **joining techniques including welding and threading**.
- Parallely **supervised** another team to maintain a continuous operation of fire resisting coating on the steel column and beam using Isolatek cement with water applying some relevant machineries.
- **Observed** the improvements of the tasks, troubleshoot problems in the field and immediate solution as early as possible.

Intern

May 2019

Caterpillar Inc. Service Center, Dhaka, Bangladesh

- Trained on **overhauling and maintenance** of natural gas and diesel engine **gensets**.
- **Inspected heavy vehicles** like excavator, wheel loader, soil compactor etc. thoroughly.
- Learned the **practical implementations** of more than **40 different hand tools** in the workshop.

MAJOR PROJECTS

Participant

Formula Kart Design Challenge Season 2

Nov 2017-Mar 2018

Organization: Fraternity of Mechanical and Automotive Engineers - FMAE, India

Funded by: Dept. of ME, Rajshahi University of Engineering & Technology

- **Designed & constructed a go-kart** was with a 125CC gasoline engine and a chain power transfer system.
- Collected different parts from the local market and applied **drilling, grinding, lathe machines** from departmental machine shop while fabricating.

Lead R&D Engineers

Full Development of 225L and 50L Refrigerator models and Cooling Development of 365L, 358L, 348L, 333L, 312L, 309L, 307L, 295L, 213L and 193L (2 versions)

Nov 2021–Present

Organization: Walton

- Led the **complete mechanical and cooling design cycles** of refrigerator systems using SolidWorks, FEA and CFD-based thermal analysis, optimizing heat exchanger layout and thermal pathways for improved system efficiency and reduced refrigerant charge.
- Provided end-to-end **IoT and control-system logic** requirements, enabling seamless microcontroller implementation (STM32/ARM/AVR) across refrigerator platforms.

CO-CURRICULAR ACTIVITIES

Joint Secretary (Technical)

2017–2021

Society of Automotive Engineers, Dept. of ME, RUET

- **Organizing member** of “Seminar on Introduction to Basic Automobile Engineering” held on 20th September, 2019 in RUET.
- **Arranging member** of “XCMG-EMSL Friendly Scholarship Program” held on 26th April, 2018 in RUET.

Development & Planning Secretary

2017–2021

RUET Career Forum, RUET

- **Coordinator** of 3rd, 4th, 5th & 6th Job Fair in 2017, 18, 19 & 21 respectively.
- **Organizing member** of other career-related seminars preached by many well-known and established personalities from different companies like Robi Axiata PLC. and RFL Group in our country.

Vice president

2017–2021

Greater Faridpur Students' Association (GFSA), RUET

- Distributed books on negligible amount of fair to the members from the association's library.
- Organizing member of fruit festival, indoor and outdoor games competition etc. for the members.

TECHNICAL SKILLS

SolidWorks, AutoCAD, ANSYS, MATLAB, MS Office, C, C++, Apps Script, AI Applications, Photopea.